

HELENA M. RICHIE

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Research focus: studying the evolution of dust and galaxies through simulation and theoretical modeling.

EDUCATION & RESEARCH

University of Pittsburgh Pittsburgh, PA 2020 – 2026
Ph.D. – Physics
Thesis: *Simulations of Dust Survival in Galactic Outflows and Connections to Observables*
Advisor: Professor Evan Schneider

University of Pittsburgh Pittsburgh, PA 2016 – 2020
B.S. – Physics & Astronomy (Honors), Mathematics

EMPLOYMENT

Flatiron Research Fellow Flatiron Institute CCA 2026 – present

Graduate Student Researcher University of Pittsburgh 2020 – 2026
Advisor: Professor Evan Schneider

Undergraduate Research Assistant University of Pittsburgh 2016 – 2020
Advisor: Professor Michael Wood-Vasey

PUBLICATIONS

First-Author Publications

10. *PAH Emission Spectra and Band Ratios for Arbitrary Radiation Fields with the Single Photon Approximation*
Richie, H. M. Hensley, B. S., ApJ, Jul. 2026, 1006, 40-51, DOI:[10.3847/1538-4357/ae7a42](https://doi.org/10.3847/1538-4357/ae7a42)
9. *Dust Evolution in Simulated Multi-phase Galactic Outflows*
Richie, H. M. Schneider, E. E., ApJ, Jan. 2026, 996, 17-38, DOI:[10.3847/1538-4357/ae12e8](https://doi.org/10.3847/1538-4357/ae12e8)
8. *Dust Survival in Galactic Winds*
Richie, H. M. Schneider, E. E. Abruzzo, M. W. Torrey, P., ApJ, Oct. 2024, 974, 81-99,
DOI:[10.3847/1538-4357/ad6a1c](https://doi.org/10.3847/1538-4357/ad6a1c)
7. *Disk Instabilities Caused the 2018 Outburst of AG Draconis*
Richie, H. M. Wood-Vasey, W. M. Coban, L., Jun. 2020, JAAVSO, 48, 21-27,
DOI:[10.48550/arXiv.1912.01681](https://doi.org/10.48550/arXiv.1912.01681)

Co-Author Publications

6. *The Lifecycle and Emission Properties of PAHs in Cosmological Hydrodynamic Galaxy Formation Simulations*

- Naryanan, D. Torrey, P. Parente, M. Donnelly, G. Zimmerman, D. Garcia, A. **Richie, H.** Hensley, B. Marinacci, F. McKinney, J. Pope, A. Popping, G. Sales, L. Sandstrom, K. Savitch, E. Shivaie, I. Spilker, J. Whitcomb, C., Jun. 2026, [arXiv:2606.20809](#)
5. *Survival of very small carbonaceous dust grains in the inner-CGM of NGC 891 from JWST/MIRI MRS*
Chastenet, J. De Looze, I. Gordon, K. Madden, S. C. Relaño, M. Sandstrom, K. M. Van De Putte, D. Baes, M. Bianchi, S. Bolatto, A. D. Casasola, V. Dale, D. A. Duval, S. Fritz, J. Galliano, F. Glover, S. C. O. Katsioli, S. Katsis, V. Kirchschlager, F. Klessen, R. S. Levy, R. C. Meidt, S. E. Mosenkov, A. V. Nagamine, K. Pantoni, L. **Richie, H. M.** Walch, S. Williams, T. G. Xilouris, E. M., May 2026, [arXiv:2605.26864](#)
 4. *JWST Observations of Starbursts: Dust Processing in the M82 Superwind*
Cronin, S. A. Bolatto, A. D. **Richie, H. M.** Donnelly, G. P. Levy, R. C. Gordon, K. D. Tarantino, E. Boyer, M. L. Armus, L. Arens, P. A. Boogaard, L. A. Dale, D. A. Donaghue, K. Draine, B. T. Duval, S. E. Emig, K. Fisher, D. B. Glover, S. C. O. Hensley, B. S. Herrera-Camus, R. Klessen, R. S. Lai, T. S.-Y. Lenkić, L. Leroy, A. K. Lieber, A. E. De Looze, I. Lopez, S. Meier, D. S. Mills, E. A. C. Sandstrom, K. M. Schneider, E. Sheriff, K. E. Siwakoti, U. Skillman, E. D. Smith, J. D. T. Teng, Y.-H. Thompson, T. A. Tielens, A. G. G. M. Veilleux, S. Villanueva, V. Walter, F. van der Werf, P. P., May 2026, ApJ, 1002, 217-240, [10.3847/1538-4357/ae5f66](#)
 3. *Effects of Numerical Resolution on Simulated Cloud-Wind Interactions*
Leary, H. J. **Richie, H. M.** Schneider, E. E. Oct. 2025, ApJ, 998, 281
 2. *Resolution Dependence of Cloud–Wind Simulations*
Leary, H. J. Schneider, E. E. **Richie, H. M.** Nov. 2024, RNAAS, 8, 286
DOI:[10.3847/2515-5172/ad9256](#)
 1. *JWST MIRI and NIRCам observations of NGC 891 and its circumgalactic medium*
Chastenet, J. De Looze, I. Relano, M. Dale, D. A. Williams, T. G. Bianchi, S. Xilouris, E. M. Baes, M. Bolatto, A. D. Boyer, M. L. Casasola, V. Clark, C. J. R. Fraternali, F. Fritz, J. Galliano, F. Glover, S. C. O. Gordon, K. D. Hirashita, H. Kennicutt, R. Nagamine, K. Kirchschlager, F. Klessen, R. F. Koch, E. W. Levy, R. C. McCallum, L. Madden, S. McLeod, A. F. Meidt, S. E. Mosenkov, A. V. **Richie, H. M.** Saintonge, A. Sandstrom, K. M. Schneider, E. E. Sivkova, E. E. Smith, J. D. T. Smith, M. W. L. van der Wel, A. Walch, S. Walter, F. Wood, K. Oct. 2024, A&A, 690, A348
DOI:[10.1051/0004-6361/202451033](#)

FELLOWSHIPS & AWARDS

- **Outstanding Poster Award**, Aromatic Universe Conference, Summer 2026
- **Z. Daniels Fellowship**, University of Pittsburgh, Spring 2026
- **PITT PACC Fellowship**, University of Pittsburgh, Spring 2025
- **Outstanding Oral Presentation**, University of Pittsburgh Grad Expo, Mar. 2024
- **Arts and Sciences Predoctoral Fellowship**, University of Pittsburgh, Summer 2021
- **NASA Pennsylvania Space Grant** (*funding for undergraduate research*), 2017, 2018, 2019, 2020

PROFESSIONAL DEVELOPMENT

- **Kavli Summer Program in Astrophysics** Jun.–Aug. 2025, Charlottesville, VA, USA

- **International High Performance Computing Summer School** Jul. 2024, Kobe, Japan
- **Argonne Training Program for Extreme Scale Computing** Jul. 2023, Chicago, IL, USA

SELECTED PRESENTATIONS

Invited Talks

- Hot Topics Seminar, University of Toledo, Nov. 2025
- CCAPP Seminar, The Ohio State University, Oct. 2025
- AstroLunch Seminar, University of Pittsburgh, Sept. 2025
- Astronomy Tea Talk, Indiana University Bloomington, Mar. 2025
- AAVSO 109th Annual Meeting, Virtual Event, Nov. 2020

Contributed Conference Talks and Department Seminars

- TASTY Talk, University of Toronto, Nov. 2025
- The Fifth Pan Dust Conference, Tucson, AZ, USA, Nov. 2025
- ITC Lunch Talk, Harvard University, Oct. 2025
- CCA Galaxy Formation Seminar, The Flatiron Institute, Oct. 2025
- CIERA Theory Group Meeting, Northwestern University, Sept. 2025
- The Physics and Impact of Astrophysical Dust Conference, Aspen Center for Physics, Feb. 2024
- Illuminating the Dusty Universe: A Tribute to the Work of Bruce Draine, Florence, Italy, Nov. 2023
- Conference for Undergraduate Women in Physics, Carnegie Mellon University, Jan. 2020

Posters

- Exploring the Aromatic Universe in the JWST Era, London, ON, Canada, Jul. 2026
- Evolution of Dust and Gas throughout Cosmic Time, Hiroshima, Japan, Dec. 2024
- Oak Ridge Leadership Computing Facility User Meeting, Knoxville, TN, USA, Sept. 2024
- IAU Symposium 377 on Galaxy Formation, Kuala Lumpur, Malaysia, Feb. 2023
- Oak Ridge Leadership Computing Facility User Meeting, Knoxville, TN, USA, Sept. 2023
- 235th Meeting of the American Astronomical Society, Honolulu, HI, USA, Jan. 2020
- The 2019 Quadrennial Physics Congress, Providence, RI, USA, 2019
- Emerging Researchers in Exoplanet Science V, Cornell University, Jul. 2019
- Emerging Researchers in Exoplanet Science IV, The Pennsylvania State University, Jul. 2018

TECHNICAL & PROFESSIONAL SKILLS

Software Development:	Cholla (<i>developer</i>), pah_spec (<i>primary code architect</i>)
Programming Languages:	C, C++, Python, Bash
Packages & APIs:	CUDA, HIP, MPI, OpenMP, NumPy, Matplotlib, Astropy
Software Tools:	Clang Tools, Make, HDF5, FITS, Slurm
HPC Systems Used:	OLCF: Summit, Andes, & Frontier, ALCF: Theta, University Clusters

TEACHING

Classes

- **ASTRON 0089 Stars, Galaxies, and the Cosmos**

Fall 2020, Spring 2021

Teaching Assistant. Taught recitation, held office hours, proctored exams, and graded homework assignments and exams for ~100 students.

– **ASTRON 1263 Techniques of Astronomy Lab**

Fall 2020

Teaching Assistant. Assisted with labs, held office hours, and assisted with grading for all (~ 20) students.

Workshops and Tutorials

- AstroPGH-TAMU Python Boot Camp, *planned and taught session on Python Basics*, Jun. 2024
- Pitt Astrophysicist for a Day, *planned lecture and coding activity, “Connecting Galaxy Morphology and Large-Scale Structure,”* Apr. 2023
- AstroPGH-TAMU Python Boot Camp, *planned and taught session on Advanced NumPy*, Jun. 2023
- Pitt Provost Academy, *planned and taught multi-day observing workshop for incoming undergraduates, “Introduction to Observational Astronomy and Transiting Extrasolar Planets,”* Sept. 2019

SERVICE & OUTREACH

Organizations

- *Astronomy on Tap Pittsburgh*, **organizer** and **co-host** (2023 – 2026)
- *Pitt Grad Workers Union Organizing Committee*, **member** (2023 – 2026)
- *Pitt Physics & Astronomy Direct Action Coalition*, **member** (2023 – 2026)
- *American Physical Society Adopt-A-Physicist*, **mentor** (2018, 2021, 2022, 2023, 2025)
- *Pitt Society of Physics Students Mentoring sUpporting, and cOnnecting studeNts*, **mentor** (2019)
- *Survey of Transiting Extrasolar Planets at the University of Pittsburgh*, **Lead Undergraduate** (2017 – 2020)
- *Pitt Society of Physics Students*, **member** (2016 – 2020)

Public Talks and Events

- *Astronomy on Tap Pittsburgh*, Pittsburgh, PA, (2026)
- *Allegheny Observatory Public Lecture*, Pittsburgh, PA, (2026)
- *Plum Senior High School Career Day*, **speaker** (2024, 2025)
- *Pitt Astrophysicist for a Day*, **panelist** (2023)
- *Astronomy on Tap Pittsburgh*, **speaker** (2023)
- *The Ellis School Science Class Visits*, **speaker** (2023, 2024)
- *Norwin Senior High School Science Alumni Day*, **speaker** (2017, 2018, 2019, 2021, 2022, 2023, 2025)